

QC analysis of raw sequencing data

Meryem Stroosma

2025-12-12

Content

- Introduction
- Research question
- Packages and requirements
- Obtaining the data
- Analysis with FastQC
- Analysis with multiQC
- Conclusion
- Discussion

Introduction

In this RMarkdown the FastQC and multiQC analysis of the .fastq-files from [GSE138852](#) is described. These .fastq-files contain the raw sequencing data and the phred-scores generated on the Illumina NextSeq 500 high-output lane with the 10x Genomics Chromium Single Cell 3' Library & Gel Bead Kit v2. This markdown explains how analysis should be done and what the conclusion and discussion is in regards to the research question.

Research question

For this project the main research question is *“How can a reliable and reproducible analysis pipeline be established to identify cell-type-specific differential gene expression from snRNA-seq data in Alzheimer’s disease?”*, the first step in answering this question is looking at the quality of sequencing data. The sub-question that is answered in this RMarkdown is as followed: Do the sequencing reads have a high enough Q-score to further analyse the data ($Q \geq 30$)?

Packages and Requirements

In order to run the scripts for QC-analysis a few packages need to be installed first, for analysis an environment was created in miniconda with the command line prompt;

```
conda create -n snrnaseq
```

The conda environment was activated with the prompt

```
conda activate snrnaseq
```

Install the required packages in the conda environment. For obtaining the fastq-files and analyses the following packages are required:

- sra-tools
- fastqc
- multiqc

Run the following commands in the terminal to install the packages:

```
conda install bioconda::sra-tools
conda install bioconda::fastqc
conda install bioconda::multiqc
```

Obtaining the data

The first step in the data-analysis pipeline is obtaining the data. The accession numbers of the GSE138852 dataset that need to be downloaded are listed in the text file *SRR_Acc_List.txt*. To download all the fastq files a bash script is written (*01_download_SRR_fastq.sh*), this script uses the *sra-tools*-package. The first section of this script defines the location where the *SRR_Acc_List.txt* is stored and an output directory, if this is non existing the script will create the output directory prior to fetching the files. After creating an output directory the script uses the *SRR_Acc_List.txt* and reads the first accession number that starts with SRR, it downloads the accession number with *fastq-dump* from the GEO database and stores it in a compressed format (gzip) to the output directory. After downloading a file it searches for the next accession number in the textfile and repeats the cycle.

The script takes a long time since the downloads are quite large, therefore it is recommended to add *nohup* and *&* to the command line to run the script on the background of the terminal. Navigate to the scripts folder and run this script from the terminal with the following command:

```
nohup bash 01_download_SRR_fastq.sh &
```

```
#!/bin/bash
# Script to download .fastq-files from a list with SRR ID's with fasterq-dump

# Install packages
# conda install bioconda::sra-tools

# Stop when errors occur
set -euo pipefail

# Path to directories
project="$HOME/ad_transcriptomics_dge"
SRR_list="$project/raw_data/SRR_Acc_List.txt"
output="$project/raw_data/fastq_data"

# Create output directory
mkdir -p "$output"

while read -r SRR || [[ -n "$SRR" ]]; do
    [[ -z "$SRR" ]] && continue # skip empty lines
    echo ">>> Downloading $SRR ..."
    fastq-dump "$SRR" --gzip -O "$output"
    # Result: $output/${SRR}.fastq.gz
done < "$SRR_list"

echo "Files are saved in $output"
```

Analysis with FastQC

For FastQC analysis the downloaded fastq-files are used as input data. The fastq files contain the sequencing reads and the quality score (Q-score or Phred score) that are associated with those reads. For each read there are four lines:

1. The first line is the identifier, this is the ID that is generated by the sequencer.
2. The second line contains the nucleotide sequence.
3. The third line is the symbol "+", which separates the sequence from the Q-score
4. The fourth line contains the Q-score this is given as a character from the ASCII table.

Analysis of the fastq files is done using a script (02_fastqc.sh) that analyzes all fastq files one after the other. This script uses the fastqc-package to create a report, the output of this script are two files:

- A html report, which contains a variety of plots such as the per base sequence quality and per sequence quality scores.
- A zipfile that contains the raw data for each plot in text format.

To run this script it is recommended to add nohup and & to the command line since there are quite a lot of files to analyse. Navigate to the scripts folder and run this script from the terminal with the following command:

```
nohup bash 02_fastqc.sh &
```

```
#!/bin/bash
# This script performs QC control on .fastq files

# Stop when errors occur
set -euo pipefail

# Activate conda environment
source ~/miniconda3/etc/profile.d/conda.sh
conda activate snrnaseq

# Install packages
# conda install bioconda::fastqc

# Path to directories
project="$HOME/ad_transcriptomics_dge"
input_dir="$project/raw_data/fastq_data"
output_dir="$project/analyses"

# Create output directory
mkdir -p "$output_dir"

# For-loop for FastQC analysis on each .fastq.gz-file
for fastq_file in "$input_dir"/*.fastq.gz
do
    basename=$(basename "$fastq_file" .fastq.gz)
    echo "Running FastQC analysis on $basename."

    fastqc "$fastq_file" \
        --memory 4096 \
        -o "$output_dir"
done
echo "FastQC analysis completed."
```

The html report will automatically be saved to the output folder that was specified in the script, two examples of the plots that were analysed were visualized in figure 1 and figure 2.

In figure 1 an example is shown of the plot that is generated for the per base quality score. On the x-axis the position of the base in the read and on the y-axis the quality score is shown. The colours on the background of the plot are an indication of the quality, with green being a very reliable quality, orange is of less reliable quality and red being of poor quality. The blue line in this plot is the mean quality for each position in the read. In this figure we can also see that a boxplot is generated, with the yellow boxes being the interquartile range (25th to 75th percentile) for that position in the read, and the outer black lines indicate the 10th and 90th percentile.

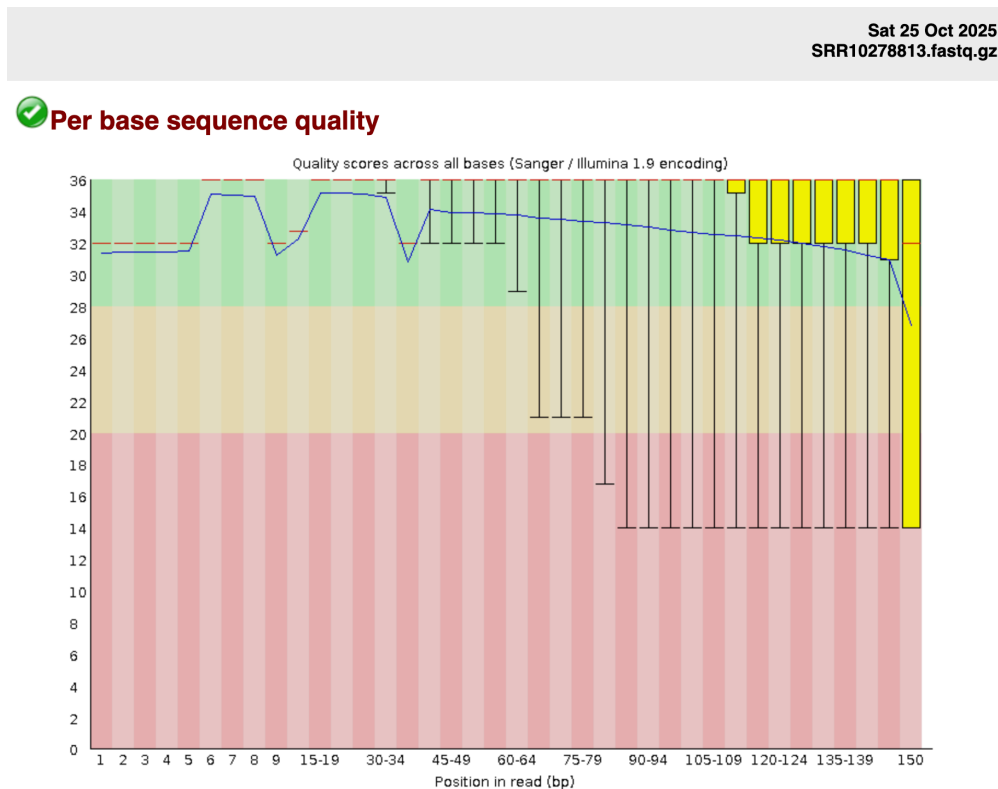


Figure 1: Per base quality score. This figure shows the per base quality score of the data of accession number SRR10278813. This plot was generated with the fast QC package.

In figure 2 an example is shown of the plot that is generated for the per sequence quality score. On the x-axis the mean quality score is shown and on the y-axis the number of sequences with that quality score. If most of the sequences in this plot have a high quality score it usually means that the sequencing run is successful.

✓ Per sequence quality scores

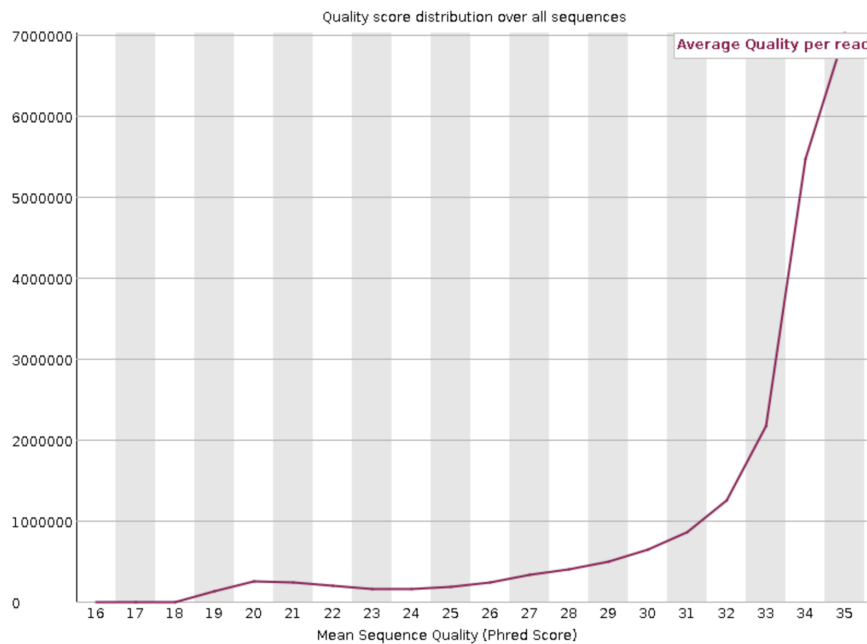


Figure 2: Per sequence quality score. This figure shows the per sequence quality score of the data of accession number SRR10278813. This plot was generated with the fast QC package.

Analysis with MultiQC

The analysis of all the plots generated in **analysis with FastQC** is quite time-consuming. To simplify the process and enable the evaluation of all samples simultaneously, the package MultiQC was used. By using MultiQC it is no longer necessary to inspect each FastQC report individually, since the tool bundles all FastQC outputs into a single interactive report. No script was required for MultiQC; after installation it can be run directly from the command line. To perform a MultiQC analysis, it is important to first navigate to the directory containing the FastQC reports, as MultiQC uses these files as its input.

```
cd path/to/fastqc_reports
multiqc .
```

After running MultiQC, a report is generated in the directory where the analysis was performed. This report contains the same plots that a FastQC report produces; however, a MultiQC report provides interactive plots in which all input samples are layered on top of one another, making it easier to compare data across runs and samples. Figure 3 shows the plot for per-base quality scores. Unlike the FastQC output, this version does not include boxplots; instead, each line represents a different accession number. In the HTML version of the MultiQC report, the plot is interactive—when hovering over a line, the corresponding accession number, read position, and mean Phred score are displayed.

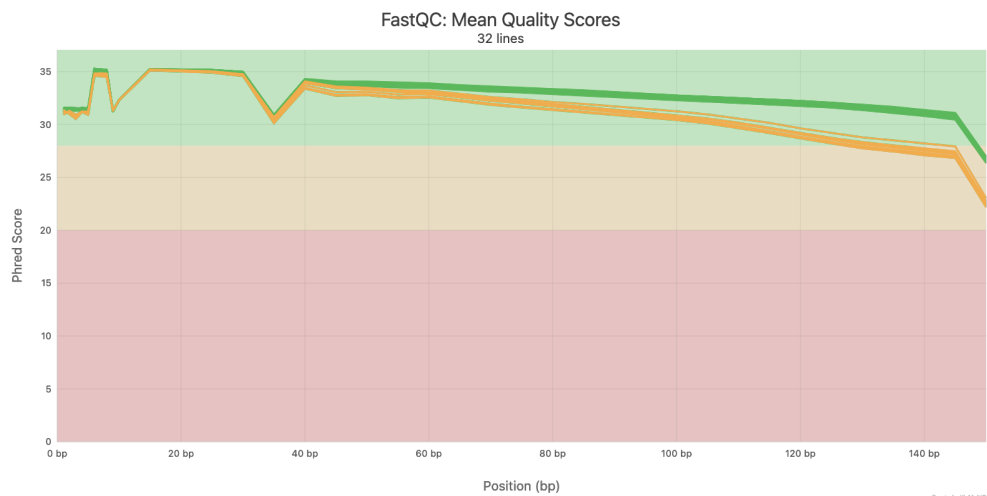


Figure 3: Per base quality score. This figure shows the per base quality score of all of the data from GSE138852. This plot was generated with the multiQC package.

Conclusion

For all the samples the average Phred scores remain above acceptable thresholds ($Q > 30$), at the end of the reads a slight incline is observed, however this is expected when sequencing with Illumina. Based on the average Phred scores it can be concluded that basecalling is performed with high accuracy.

When looking at the Per Sequence Quality Scores, it can be observed that the curve for the average Q-score of individual reads shows a peak in the green section of the plot ($Q > 30$). This indicates that the majority of the reads are of very good quality and that the dataset appears to be reliable based on this information.

The plot for GC content shows a peak as expected around 41% since human DNA has a GC content about 41% [1](#). There is a small shoulder in the peak at around 80%.

The overall conclusion is that the data can be used for further processing, the quality of sequencing runs looks good with an average Q-score of >30 . The GC-content is as is expected for the organism and the drop in quality at the end of the reads is a known pattern for illumina.

Discussion

In general all the average Phred-scores of the different samples are of high quality, however at the end of each read the quality starts to go down. In order to only use reads with $Q>30$, the end of the reads could be trimmed. Another option is to completely filter out everything with $Q<30$. Since the profile of the plot is typical for Illumina, the choice was made to keep the data as is for now. Also filtering out data could potentially reduce the resolution of the dataset.

The plot for GC-content showed a small shoulder at 80%, this could be a small contamination but since the shoulder shows at the same point in all the samples this seems unlikely. It might also be that there are some GC-rich genomic regions or repetitive elements present [2](#).